

Chapter 124

Modifiers

This chapter details the modifiers available in Fusion.

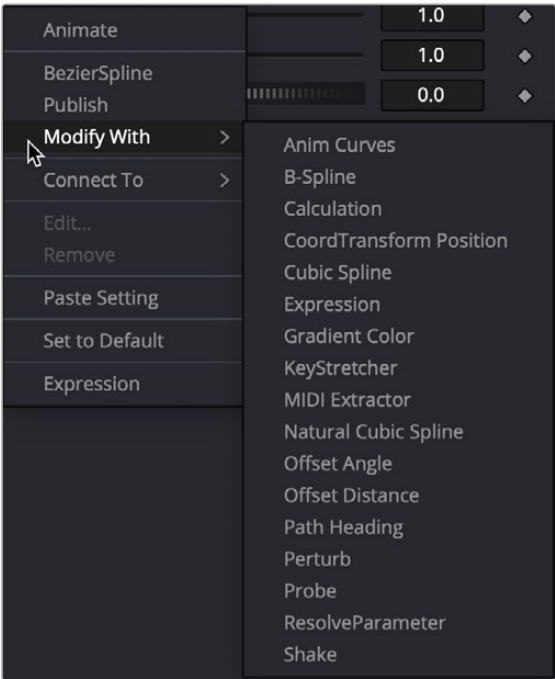
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Modifiers

Modifiers are extensions to a node’s standard set of parameters found in the Inspector; in fact, modifiers are designed to control other parameters. They can be as simple as a motion path or linking two parameters. However, they can also be elaborate expressions, procedural functions, external data, third-party plugins, or scripted Fuses. You can add modifiers by right-clicking over a parameter in the Inspector and choosing a modifier from the menu. Alternatively, you can right-click a control in the viewer. Not all modifiers are displayed in the right-click menu for all parameters. Some modifiers work only on specific parameter types.

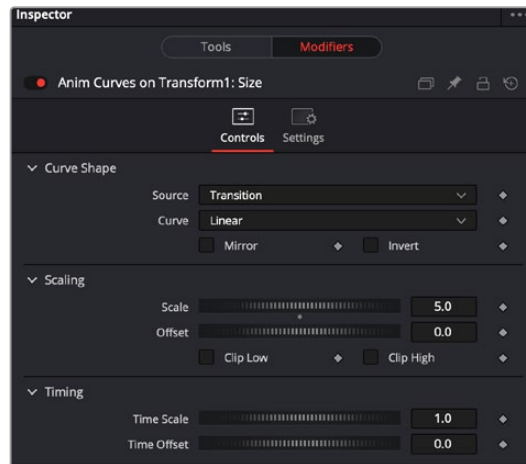
NOTE: Text3D and Text+ have additional text-specific modifiers, which are covered in their nodes’ sections.



Modifiers right-click menu

Anim Curves

The Animation Curves modifier (Anim Curves) is used to dynamically adjust the timing, values, and acceleration of an animation, even if you decide to change the duration of a comp. Using this modifier makes it infinitely easier to stretch or squish animations, create smooth motion, add bouncing properties, or mirror animations curves without the complexity of manually adjusting splines.



The Inspector for the Anim Curves modifier

When creating Fusion templates for the Edit and Cut page in DaVinci Resolve, the Anim Curves modifier allows the keyframed animation you've created in Fusion to stretch and squish appropriately as the transition, title, or effect's duration changes on the Edit and Cut page Timelines.

Curve Shape Controls

The controls for the Anim Curves modifier appear in the modifier's tab of the Inspector. The Curve Shape controls determine the acceleration or shape of the animation curve.

- **Source:** This drop-down menu has three options based on how the comp is created from DaVinci Resolve's Edit page.

Transition: This setting is automatically selected when the comp is created from an Edit page transition effect. If the duration of the transition is updated in the Edit page, the timing of the animation updates as well.

Duration: Use this setting when the comp is created from a clip on the Edit page. The animation timing will update if the clip's duration changes by trimming.

Custom: Displays an Input dial to manually control the timing.

- **Input:** This dial is only visible when Source is set to Custom. It is used to change the input keyframe value.
- **Curve:** The Curve drop-down menu selects the interpolation method used between keyframes. The three choices are: linear, easing, or custom.

Linear: The default Linear interpolation method maintains a fixed, consistent acceleration between keyframes.

Easing: Displays interpolation menus for both the start of the curve (In) and the end of the curve (Out).

Custom: Opens a mini Spline Editor to customize the interpolation from the start of the animation to the end.

- **Mirror:** Plays the animation forward, and after reaching the end, it returns to the starting value. This causes the initial animation to be twice as fast, since the second half of the comp is used for the reverse animation.
- **Invert:** Flips the animation curve upside-down so that the values start high and end low.

Scaling

The Scale parameters modify the animation values using relative adjustments.

- **Scale:** This number is a multiplier applied to the value of the keyframes. If the Scale value is 2 and a keyframe has a value of 0, it remains 0. If the Scale value is 2 and a keyframe has a value of 10, the result is as if the keyframe is set to 20. This can be thought of as the ending value for the animation. It is best to set this while viewing the last frame in the comp.
- **Offset:** The offset is added to the keyframe values and can be thought of as the starting value for the animation. It is best to set this while viewing the first frame in the comp.
- **Clip Low:** Ensures the output value never dips below 0.0.
- **Clip High:** Ensures the output value never exceeds 1.0.

Timing

The Timing parameters adjust the animation timing using relative values.

- **Time Scale:** Stretches or squishes the animation, causing it to run faster or slower. A value of 1.0 keeps the animation running for the comp's duration (unless you have customized the animation using other controls in the Modifier).
- **Time Offset:** This value delays the animation as a fraction of its total duration. A value of 0.0 applies no delay. A value of 0.5 delays the animation starting point midway into the comp's duration.

Using the Anim Curves Modifier to Create a Custom Transition

To understand how to use the Anim Curves modifier for a transition, let's create a simple scaling dissolve.

- 1 Add a standard cross dissolve in the Edit page Timeline.
- 2 Right-click over the transition and choose Convert to Fusion Cross Dissolve.
- 3 Right-click over the transition again and choose Open in Fusion page.
- 4 Add a Transform node to Medialn1 and to Medialn2.
- 5 Select the Transform node attached to the Medialn2.
- 6 In the Inspector, right-click the Size control, then choose Modify With > Anim Curves from the contextual menu. Adding this modifier to the Size control will cause the slider to animate from 0 to 1 for the Cross Dissolve's duration.
- 7 Select the Transform node attached to the Medialn1.
- 8 In the Inspector, right-click the Size control, then choose Modify With > Anim Curves from the contextual menu.
- 9 At the top of the Inspector, click the Modifier tab and click the Invert button. Inverting the animation curves causes Medialn1 to scale opposite of Medialn2.
- 10 In the Modifiers tab, set the Curve drop-down menu to Easing, and experiment with the different ease-in ease-out curve types from the In/Out drop-down menus.

Once you create a Macro from this node tree and save it as a Transition template, you can apply it in the Edit page Timeline. If you change the transition duration in the Edit page, the animation timing will update appropriately.

Using the Anim Curves Modifier with Paths

To understand how to use the Anim Curves modifier's with a Path modifier, let's use the premise that you want to create text that falls from the top of the frame and bounces as it reaches the bottom of the frame.

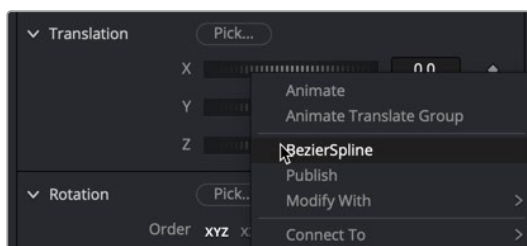
- 1 In Fusion, create two keyframes that cause text to start at the top of the frame and drop to the bottom. This automatically creates a Path modifier.
- 2 In the Inspector's Modifier tab, right-click over the Displacement parameter and choose Insert > Anim Curves. The animation is normalized to the duration of the comp.
- 3 Set the Source menu to Duration, since this is not a transition and we are not customizing the duration.
- 4 From the Curve menu, choose Easing, then for the Out menu, choose Bounce.
- 5 Play the animation to see the Bounce animation.
- 6 To make the bounce occur halfway down the frame, change the Scale to .05.
- 7 To make the animation run twice as fast, enter 2.0 in the Time Scale parameter.

Once you create a macro from this node tree and save it as a Title template, you can apply it in the Edit page Timeline. If you change the title's duration in the Edit page, the animation timing will update appropriately.

TIP: To view the resulting animation curve in the Spline Editor, select the parameter name in the Spline Editor's header. The spline is updated as you change the controls.

Bézier Spline

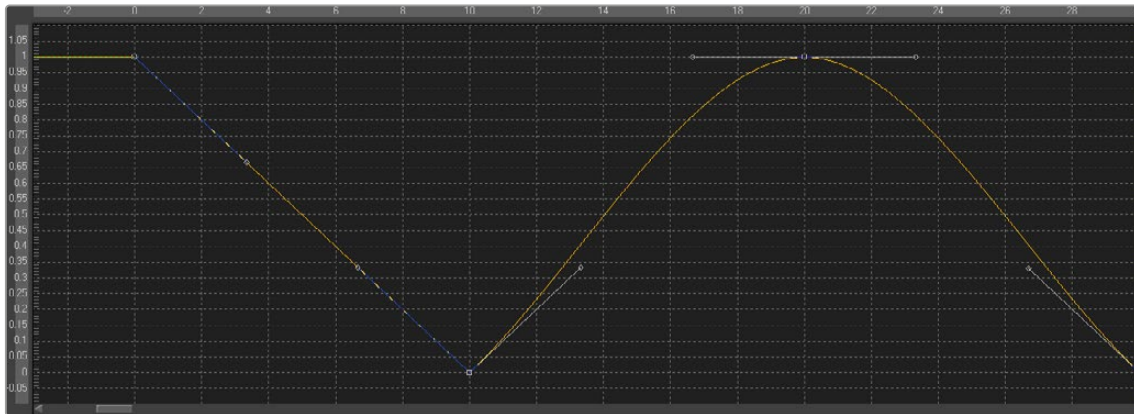
The Bézier Spline is one of the animation modifiers in Fusion and is typically applied to numerical values rather than point values. It is automatically applied when you keyframe a parameter or each time you right-click a number field and select Animate.



Bézier Spline modifier menu

Usage

You can add the Bézier Spline to the Spline Editor by right-clicking a number field and selecting BezierSpline. Since this is the most common choice for animation splines, it is separated from the Modify With menu for quicker access. Selecting BezierSpline from the menu adds a keyframe at the current location and displays a Bézier Spline in the Spline Editor.



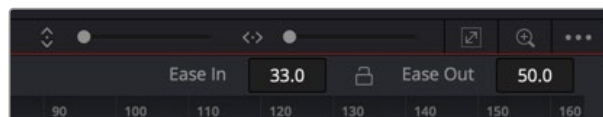
Bézier Spline Editor

Unlike most modifiers, this modifier has no actual Controls tab in the Inspector. However, the Spline Editor displays the Bézier Spline, and it can be controlled there. The Bézier Spline offers individual control over each control point's smoothness using Bézier handles. The smoothness is applied in multiple ways:

- To make the control points smooth, select them, and press Shift-S. The handles can be used to modify the smoothness further.
- To make the control points linear, select them, and press Shift-L. These operations can also be performed using the contextual menu.
- Select the control point(s), right-click, and select Smooth or Linear. The menu also allows the user to smooth a spline using a convolution analysis called a Savitzky-Golay filter. Select the control point(s), right-click, and select Smooth Points -Y Dialog.

Ease In/Out

Traditional Ease In/Out can also be modified by using the number field virtual sliders in the Spline Editor. Select the control points you want to modify, right-click, and select Ease In/Out... from the contextual menu. Then use the number field virtual sliders to control the Ease In/Out numerically.

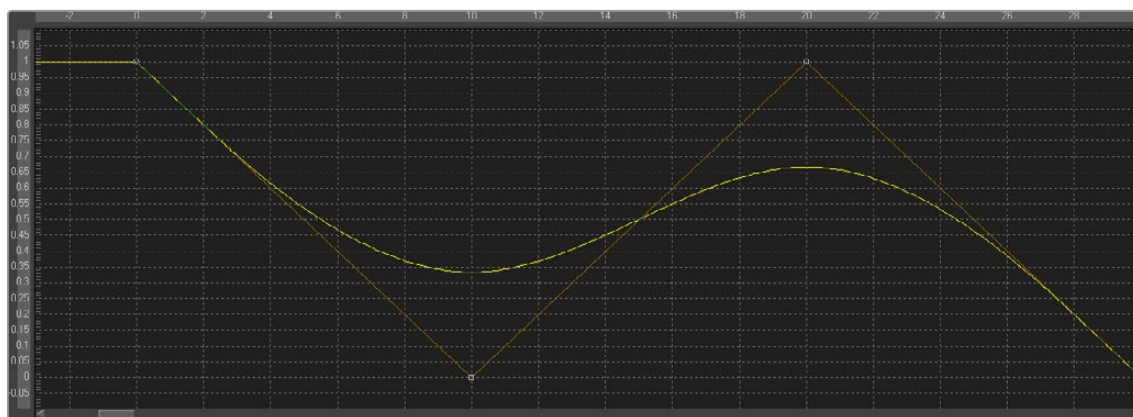


Spline Ease In/Out modifier

B-Spline

An alternative to the Bézier Spline, B-spline is another animation modifier in Fusion and is typically applied to numerical values rather than point values. It is applied by right-clicking a parameter and selecting Modify With > B-Spline.

Usage



B-Spline Editor

- This animation spline modifier has no actual Controls tab. However, the Spline Editor displays the B-spline, and it can be controlled there. Notice that, though the actual value of the second keyframe is 0, the value of the resulting spline is 0.33 due to the unique smoothing and weighing algorithms of a B-spline.
- The weight can be modified by clicking the control point to select it, holding the W key, and moving the mouse left and right to lower or increase the tension. This is also done with multiple selected control points simultaneously.

Calculation

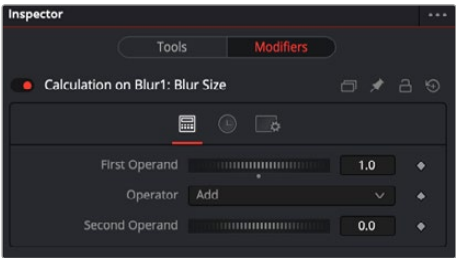
Calculations are used to create indirect connections between parameters. A Calculation can perform a mathematical expression based on two operands, where each operand can be connected to another parameter or set manually.

Additionally, using Time offsets and Scale controls in the Time tab, the Calculation control can access values of a parameter at times other than the current time.

The Calculation's most common use is for connecting two parameters when one value range or scope is inappropriate for the other parameter.

NOTE: The Expression modifier is essentially a more flexible version of the Calculation modifier, with a single exception. It is far easier to manipulate the timing of the operands provided in the Calculation modifier than it is to do so with an Expression.

Inspector



Calculation Calc tab

Calc Tab

The Calc tab includes two dials used for the connected parameter and value that gets mathematically combined. The Operator menu selects how the Second Operand value combines with the parameter’s value.

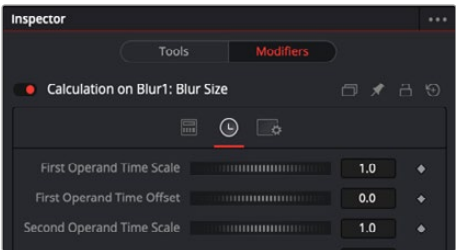
First and Second Operand

These sliders are connected to published or animated parameters or manually set to the desired values for the calculation.

Operator

Select from the mathematical operations listed in this menu to determine how the two operands are combined. Clicking the drop-down arrow opens the menu with the following options:

- Add
- Subtract (First - Second)
- Multiply
- Divide (First / Second)
- Divide (Second / First)
- Subtract (Second - First)
- Minimum
- Maximum
- Average
- First only



Calculation Time tab

Time Tab

The Time tab is used to modify the time of the Calculation modifier. The controls here retime the speed of the effect or offset it in time.

First and Second Operand Time Scale

These sliders multiply the frame number and return the value of the operands at the multiplied frame number. A value of 1 returns the value of the operand at frame x when the composition is on frame x. For example, if the first operand is animated with a value of 1 to 50 from frame 0 to 10, then a scale of 0.5 would cause the calculation to return a value of 25 at frame 10 (effectively slowing the animation by half for the purposes of the calculation).

First Operand and Second Operand Time Offset

These sliders return the value of the operand at the Time Offset specified. A value of 10 would return the value of the operand 10 frames forward in time, and -10 would return the value of the operand 10 frames back in time. See the example below for a practical example.

Example

The following example uses a calculation to apply blur to a Text node in inverse proportion to the size of the text.

- Open a new composition that starts on frame 0 and ends on frame 100.
- At frame 0, add a Text+ node to the composition.
- Enter a small amount of text and set the size to 0.05
- Click the Keyframe button to the right of the Size slider to add a keyframe.
- Move to frame 100 and set the Size value to 0.50.
- Connect a Blur node after the Text+ node.
- View the Blur node in one of the viewers.

To have the blur decrease in strength as the text gets bigger, a simple “pick whip”-style parameter linking does not work. The controls cannot be directly connected together because the values of the Text Size control are getting bigger instead of smaller.

- Right-click the Blur Size and select Modify With > Calculation from the contextual menu.

This adds a Calculation modifier to the Blur node. At the top of the Inspector, a new set of controls appears in the Modifiers tab while the Blur node is selected.

- At the top of the Inspector, select the Modifiers tab (F11).
- Right-click the First Operand slider and select Connect To > Text 1 > Size from the contextual menu.

Although the Blur Size is now connected to the Text Size parameter, this connection isn't very useful. The maximum value of the Blur Size control is 0.5, which is hardly noticeable as a blur.

- Set the Operator drop-down menu to Multiply.
- Set the Second Operand slider to 100.
- Now the first operand is multiplied by 100, and adjusting the dial gives you a much blurrier blur.
- Switch to the Time tab of the modifier and set the First Operand Time Scale to -1.0.

Normally, the first operand gets the value of the control it is connected to from the same frame as the current time. So at frame 10, the first operand is set to the same value as the Text size at frame 10. By setting this value to -1, the value is read from one frame back in time whenever the current time of the composition advances by 1 frame.

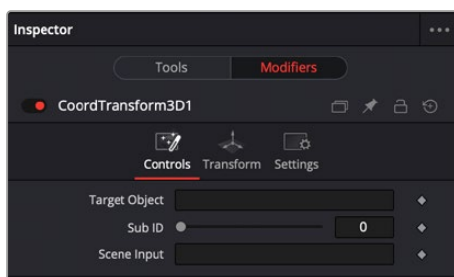
However, this means that the Calculation would be reading the value of the Text size at frame -10 when we are at frame 10 in the composition.

- To correct for this, set the First Operand Time Offset slider to 100.
- Return to the Tools tab at the top of the Inspector and press Play (Spacebar) to watch how the value of the Blur Size relates to the value of the Text Size.

CoordTransform Position

Because of the hierarchical nature of 3D in Fusion, the original position of an object in a 3D scene often fails to indicate the current position of the object. For example, an image plane might initially have a position at 1, 2, 1, but then be scaled, offset, and rotated by other tools further downstream in the node tree, ending up with an absolute location of 10, 20, 5. This can complicate connecting an object further downstream in the composition directly to the position of an upstream object. The Coordinate Transform modifier can be added to any set of the XYZ coordinate controls to calculate the current position of a given object at any point in the scene hierarchy. To add a Coordinate Transform modifier, right-click the numeric input on any node, and select Modify With > CoordTransform Position from the contextual menu.

Inspector



The Coordinate Transform modifier Controls tab

Controls Tab

The Controls tab has two fields for the target and scene input. The target is for the node continuing the original coordinates, while the scene input is used for the scene with the new coordinates.

Target Object

This control is connected to the 3D tool that produces the original coordinates to be transformed. To connect a tool, drag the node from the Node Editor into the text edit control, or right-click the control and select the tool from the contextual menu. It is also possible to type the tool's name directly into the control.

SubID

The SubID slider can be used to target an individual sub-element of certain types of geometry, such as an individual character produced by a Text 3D tool or a specific copy created by a Duplicate 3D tool.

Scene Input

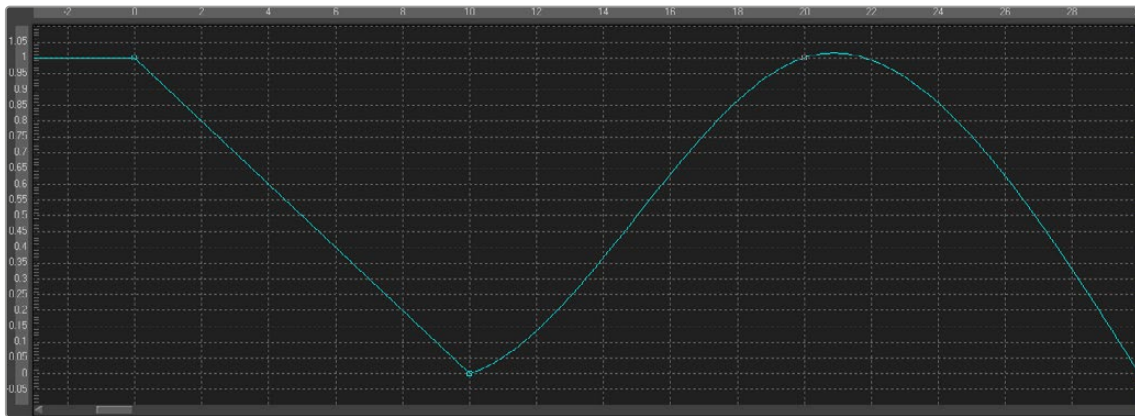
This control should be connected to the 3D tool, which outputs the scene containing the object at the new location. To connect a tool, drag and drop a tool tile from the Node Editor into the text edit control, or right-click the control and select an object from the Connect To pop-up menu.

Cubic Spline

The Cubic Spline is another animation modifier in Fusion that is normally applied to numerical values rather than point values. It can be applied by right-clicking a numerical control and selecting Modify With > Natural Cubic Spline.

Usage

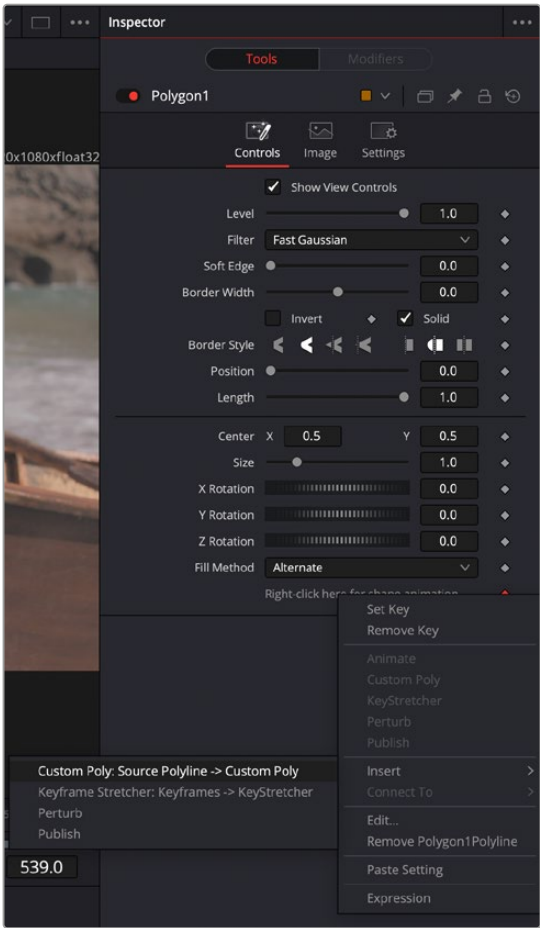
Being an animation spline, this modifier has no actual Controls tab. However, its effect can be seen and influenced in the Spline Editor.



Cubic Spline Editor

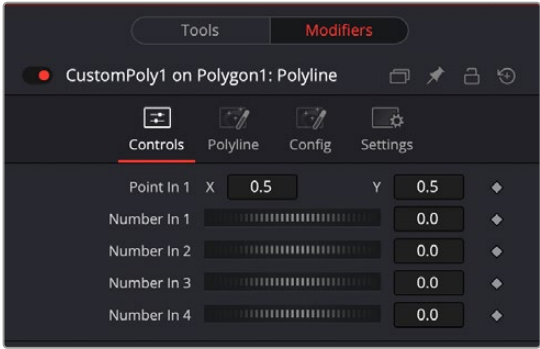
Custom Poly

The Custom Poly modifier can be added to Polygon masks or paths. Similar in function to the Custom and pCustom tools, existing points can be repositioned in the polyline, or replaced completely with a new set of points. The expressions are evaluated for each point on the output polygon. The modifier can be applied by right-clicking on the “Right-click here for shape animation” text at the bottom of the Polygon controls, and selecting Insert > Custom Poly from the contextual menu.



Selecting the Custom Poly modifier

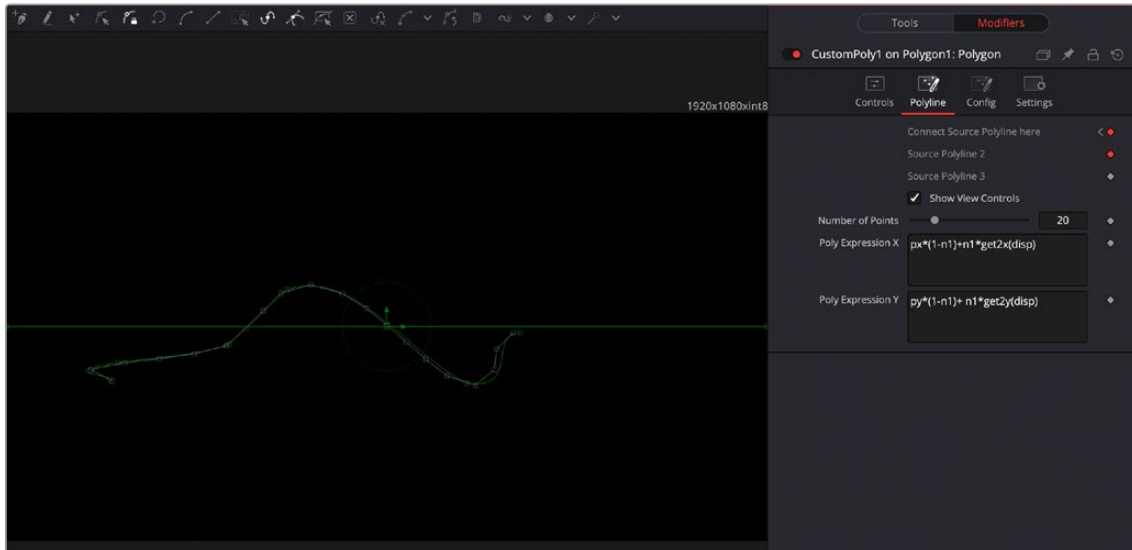
Inspector



The Custom Poly Controls tab

Controls Tab

The Custom Poly Controls tab are available once you select the modifiers tab on the Polygon's inspector. It has a single point input, and number variables to use in animating the expressions. The default is one point and four numbers, but they can be expanded to a maximum of nine. Additional points and numbers can be added from the Config tab.



The Custom Poly Polyline tab

Polyline Tab

The polyline tab exposes the controls to connect and modify the polyline.

Connect Source Polyline here

Allows you to connect other polylines to the modifier.

Show View Controls

This checkbox toggles the polyline controls visibility in the viewer.

Number of Points

The number of output points can be set, controlling the amount of custom subdivision of the polyline. A value of zero uses the number of original source points.

Poly Expression X/Y

Fields for entering the mathematical expressions. For example, $px*(1-n1)+n1*get2x(disp)$ and $py*(1-n1)+n1*get2y(disp)$.

It uses most of the same expression variables as the Expression modifier below (i.e., $n1..n9$, $p1x..p9x$, $p1y..p9y$, math functions, etc.), and it adds:

- px, py for the current point on the source polygon
- $disp$ for the point's displacement on the polyline (0.0 is start, 1.0 is end)
- $index$ for the current point's index (zero-based)
- num for the number of output points
- $getx(disp)$, $gety(disp)$ to get values from anywhere on the polyline
- $getx_at(disp, time)$, $gety_at(disp, time)$ to get values from polylines at other times
- Similarly, $get2x/y()$, $get3x/y()$, $get2x/y_at()$, $get3x/y_at()$ for the second and third source polys

Config Tab

The config tab allows you to select the amount of Points and Numbers shown in the modifier (to a maximum of 9), and to give them custom names for easier organization.

Expression

An expression is a variable or a mathematical calculation added to a parameter, rather than a straight numeric value. You can add an expression to any parameter in Fusion, or you can add the Expression modifier, which adds several tabs to the modifier Inspector. Adding this modifier to a parameter adds the ability to manipulate that parameter based on any number of controls, either positional or value-based. This modifier offers exceptional flexibility compared to the more limited Calculation or Offset modifiers, but it is unable to access values from frames other than the current time.

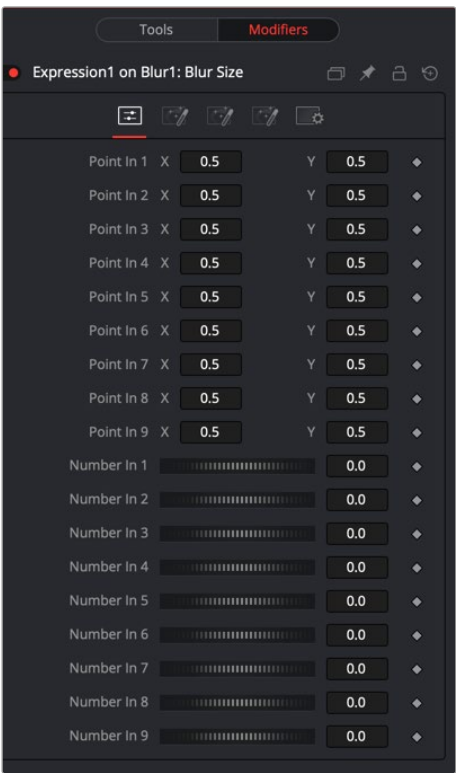
The Expression modifier accepts nine value inputs and nine position inputs that are used as part of a user-defined mathematical expression to output a value.

To add the Expression modifier to a parameter, right-click the parameter in the Inspector and choose Modify With > Expression from the contextual menu. The type of value returned by the Expression depends entirely on the type of control it is modifying.

When used with a value control (like a slider), the Expression in the Number Out tab is evaluated to create the result. When used to modify a positional control (like Center), the Point Out tab controls the result.

The Inspector's Modifiers tab contains the controls for the Expression modifier, described below.

Inspector

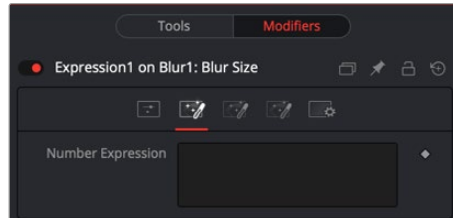


Expressions modifier controls

Controls Tab

This tab provides nine number controls and nine point controls. The values of the number controls can be referred to in an expression as n1 through n9. The X-coordinate of each point control can be referred to as p1x through p9x, while the Y-coordinate is p1y through p9y.

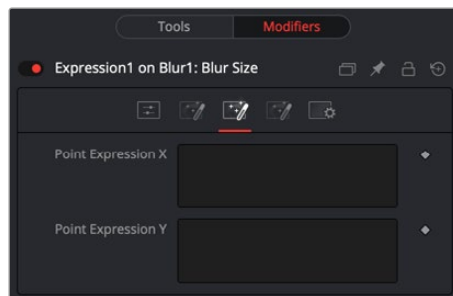
These values can be set manually, connected to other parameters, animated, and even connected to other Expressions or Calculations.



The Number Out tab

Number Out Tab

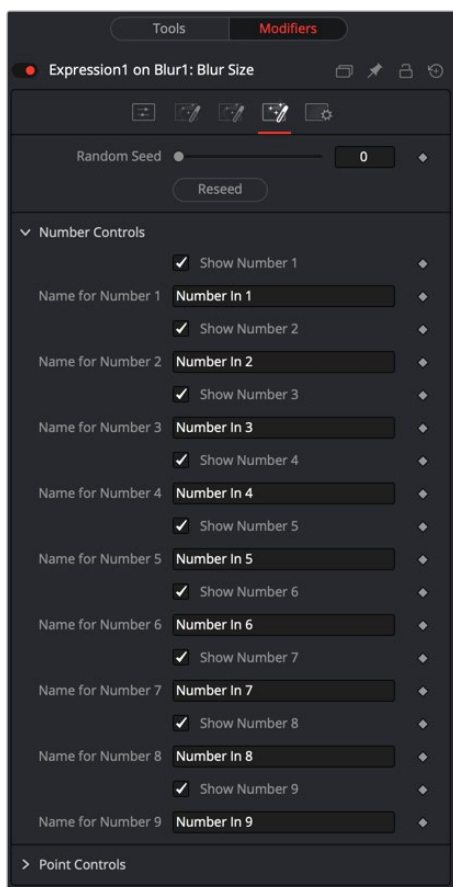
Mathematical formulas are entered using the Number In and Point In values from the Controls tab. The output modifies the control in which the Expression was applied. See below for the syntax to use in this field.



The Point Out tab

Point Out Tab

The two text boxes in this tab use mathematical formulas, accessing the Number In and Point In values from the Controls tab. The output value modifies the control in which the Expression was applied. The Expression in the top text box control is used to calculate the X-axis value, and the bottom text box is used to calculate the Y-axis control. See below for the syntax to use in this field.



The Expressions modifier Config tab

Config Tab

A good expression is reused over and over again. As a result, it can be useful to provide more descriptive names for each parameter or control and to hide the unused ones. The Config Tab of the Expressions modifier can customize the visibility and name for each of the nine point and number controls.

Random Seed

The Random Seed control sets the starting number for the Rand() function. The rand(x, y) function produces a random value between X and Y, producing a new value for every frame. As long as the setting of this Random Seed slider remains the same, the values produced at frame x are always the same. Adjust the Seed slider to a new value to get a different value for that frame.

Show Number or Point X

There are eighteen of these checkbox controls, one for each of the nine Number and nine Point inputs. Enable this checkbox to display the control for Number x or Point x in the Controls tab.

Name for Number or Point X

There are eighteen of these edit controls, one for each of the nine Number and nine Point inputs. Type a new name for the input into this edit control to assign a new name for the Input's label in the Controls tab.

Expression Syntax Formulas

Formulas are entered into the Number Out or Point Out tabs as part of an expression. They can be made up of the following functions:

n1...n9	The value of Number Input 1..9.
p1x...p9x	The X of Positional Control 1..9.
p1y...p9y	The Y of Positional Control 1..9.
time	The current time (frame number).
pi	The value of pi.
e	The value of e.
log(x)	The base-10 log of x.
ln(x)	The natural (base-e) log of x.
sin(x)	The sine of x (x is degrees).
cos(x)	The cosine of x (x is degrees).
tan(x)	The tangent of x (x is degrees).
asin(x)	The arcsine of x, in degrees.
acos(x)	The arccosine of x, in degrees.
atan(x)	The arctangent of x, in degrees.
atan2(x, y)	The arctangent of x,y, in degrees.
abs(x)	The absolute (positive) value of x.
int(x)	The integer (whole) value of x.
frac(x)	The fractional value of x.
sqrt(x)	The Square Root of x.
rand(x, y)	A random value between x and y.
rands(x, y, s)	A random value between x and y, based on seed s.
min(x, y)	The minimum (lowest) of x and y.
max(x, y)	The maximum (highest) of x and y.
dist(x1, y1, x2, y2)	The distance between point x1,y2 and x2,y2.
dist3d(x1,y1,z1,x2,y2,z2)	The distance between 3D points x1,y2,z1 and x2,y2,z2
noise(x)	A smoothly varying Perlin noise value based on x
noise2(x, y)	A smoothly varying Perlin noise value based on x and y
noise3(x, y, z)	A smoothly varying Perlin noise value based on x, y and z
if(c, x, y)	Returns x if c not 0, otherwise y.

Expression Syntax Operators

Operators are used to evaluate statements. They are combined with functions to perform logical and mathematical calculations in the Number Out and Point Out tabs.

x + y	x plus y.
x - y	x minus y.
x < y	1.0 if x is less than y, otherwise 0.0.
x > y	1.0 if x is greater than y, otherwise 0.0.
!x	1.0 if x = 0, otherwise 0.0.
-x	(0.0 - x).
+x	(0.0 + x) (effectively does nothing).
x ^ y	x raised to the power of y.
x * y	x multiplied by y.
x / y	x divided by y.
x % y	x modulo y, (remainder of (x divided by y)).
x <= y	1.0 if x is less than or equal to y, otherwise 0.0.
x >= y	1.0 if x is greater than or equal to y, otherwise 0.0.
x = y	1.0 if x is exactly equal to y, otherwise 0.0.
x == y	1.0 if x is exactly equal to y, otherwise 0.0 (identical to above).
x <> y	1.0 if x is not equal to y, otherwise 0.0.
x != y	1.0 if x is not equal to y, otherwise 0.0 (identical to above).
x & y	1.0 if both x and y are not 0.0, otherwise 0.0.
x && y	1.0 if both x and y are not 0.0, otherwise 0.0 (identical to above).
x y	1.0 if either x or y (or both) are not 0.0, otherwise 0.0.
x y	1.0 if either x or y (or both) are not 0.0, otherwise 0.0 (identical to above).

Example 1

To make a numeric control equal to the Y value of a motion path, add an expression to the desired target control and connect the Path to Point In 1. Enter the formula:

ply

into the Number Out field.

Example 2

To make the result of the Expression's Number Out be the largest of Number In 1 and Number In 2, multiplied by the cosine of Number In 3, plus the X coordinate of Point In 1, enter the formula:

```
max(n1, n2) * cos(n3) + plx
```

into the Number Out field.

Example 3

Add a Background node set to solid black and a Hotspot node. Set the Hotspot size to 0.08 and set the Strength to maximum. Modify the Hotspot center with an expression. Change the current frame to 0.

Set n1 to 0.0 and add a Bézier Spline. At frame 29, set the value of n1 to 1.0. Select both points and loop the spline using the Spline Editor. Now enter the following equations into the Point Out tab of the expression.

X-Axis Expression

```
n1
```

Y-Axis Expression

```
0.5 + sin(time*50) 4
```

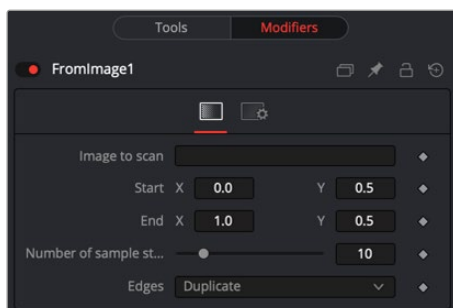
Render out a preview and look at the results. (Try this one with motion blur.)

From Image

The From Image modifier only works on gradients, like the gradient on a Background node. It takes samples of an image along a user-definable line and creates a gradient from those samples.

Unlike other modifiers, From Image is not located in the Modify With menu. This modifier can be applied by right-clicking a Gradient bar in the Inspector and selecting From Image.

Inspector



From Image modifier

Controls Tab

The From Image controls tab in the Inspector is used to select the node that contains the image you want to sample. It allows you to define the starting and ending point in the image as well as how many color samples to use in creating the gradient.

Image to Scan

Drop into this box the node from the Node Editor that you want to be color sampled.

Start X/Y, End X/Y

These two point controls define the Start and End points of the line along which the samples are taken from the image defined in the Image to Scan box.

The points can also be moved directly in the viewer.

Number of Sample Steps

This defines how many individual color samples are taken along the line. You can also see the result of this setting when you look at the actual node's Gradient control. The more sample steps you define here, the more individual color stops appear on the Gradient control. It is also possible to first create a gradient using the From Image modifier and then remove that modifier from the Gradient control again. The created gradient stays intact and can then be fine tuned by hand.

Edges

Edges determines how the edges of the image are treated when the sample line extends over the actual frame of the image to be sampled.

Black

This outputs black for every point on the sample line outside of the image bounds.

Wrap

This wraps the edges of the line around the borders of the image.

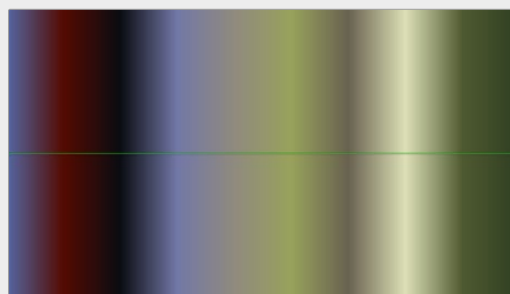
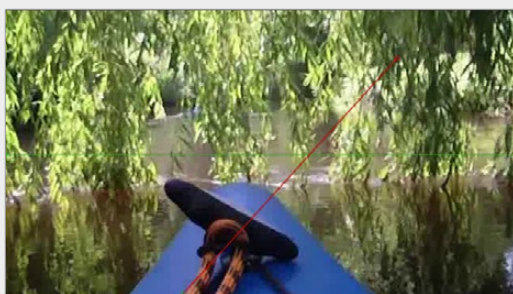
Duplicate

This causes the edges of the image to be duplicated as best as possible, continuing the image beyond its original size.

Color

This outputs a user-definable color instead of black for every point on the sample line outside of the image bounds.

Example



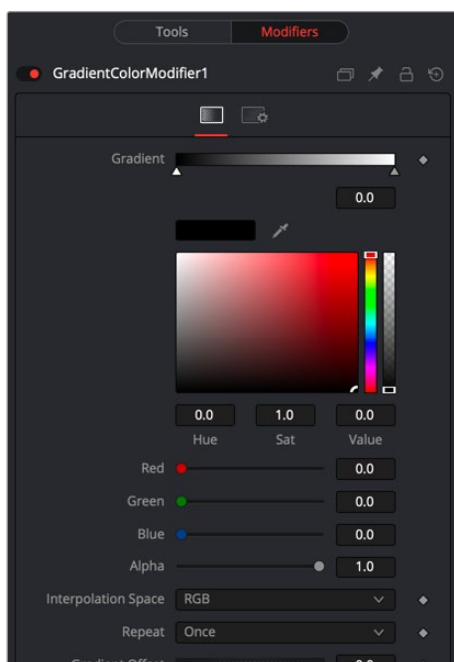
The source image on the left shows the color selection line in red. The image on the right shows the resulting gradient from that selection.

Gradient Color

The Gradient Color modifier allows you to create a customized gradient and map it into a specific time range to control a value. Use the Start and End time controls to set the frames for the animation. If the Start and End time values are set to 0, then the modifier returns the value at the starting point of the gradient. You can use the Offset control to animate the gradient manually.

It can be applied by right-clicking a parameter and selecting Modify With > Gradient Color.

Inspector



The Gradient modifier Controls tab

Controls Tab

The Controls tab consists of a Gradient bar where you add and adjust points of the gradient. Start Time and End Time thumbwheels at the bottom of the Inspector determine the time range the gradient is mapped into.

Gradient

The Gradient control consists of a bar where it is possible to add, modify, and remove points of the gradient. Each point has its color. It is possible to animate the color as well as the position of the point. Furthermore, a From Image modifier can be applied to the gradient to evaluate it from an image.

Gradient Interpolation Method

The gradient is linearly interpolated from point to point in RGB color space by default. This can sometimes result in unwanted colors. Choosing another color space may provide a better result.

Repeat

Defines how the left and right borders of the gradient are treated.



Gradients set to Once, Repeat, and Ping Pong from top to bottom, respectively, and shifting the gradient to the left.

- **Once:** When using the Gradient Offset control to shift the gradient, the border colors keep their values. Shifting the default gradient to the left results in a white border on the left; shifting it to the right results in a black border on the right.
- **Repeat:** When using the Gradient Offset control to shift the gradient, the border colors are wrapped around. Shifting the default gradient to the left results in a sharp jump from white to black; shifting it to the right results in a sharp jump from black to white.
- **Ping Pong:** When using the Gradient Offset control to shift the gradient, the border colors ping-pong back and forth. Shifting the default gradient to the left results in the edge fading from white back to black; shifting it to the right results in the edge fading from black back to white.

Gradient Offset

Allows you to pan through the gradient.

Time Controls

The Start Time and End Time thumbwheels determine the time range the gradient is mapped into. This is set in frames. The same effect can be achieved by setting the Gradient to Once and animating the offset thumbwheel.

Key Stretcher Modifier

The Keyframe Stretcher modifier is primarily used when creating title templates in Fusion for use in DaVinci Resolve's Edit page or Cut page. The Keyframe Stretcher modifier is added to an animated parameter, so the keyframes in the animated parameter stretch when the template is trimmed in the Timeline. It can be applied by right-clicking a parameter and selecting Modify with > KeyStretcher.

For more information on the Keyframe Stretcher Modifier controls, see the "Keyframe Stretcher", see *Chapter 111, "Miscellaneous Nodes,"* in the DaVinci Resolve Reference Manual or Chapter 51 in the Fusion Reference Manual.

MIDI Extractor

The MIDI Extractor modifier provides the ability to modify the value of a control using the values stored in a MIDI file. This modifier relies on some knowledge of MIDI, which is beyond the scope of this manual.

The value produced by the modifier is extracted from the MIDI event selected in the Mode menu. Each mode can be trimmed so that only specific messages for that event are processed—for example, only some notes are processed, while others are ignored. The value of the event can be further scaled or modified by additional factors, such as Scale, Velocity, Attack, and Decay.

It can be applied by right-clicking a parameter and selecting Modify With > MIDI Extractor.

Inspector



The MIDI Extractor modifier Controls tab

Controls Tab

The Controls tab is used to load the MIDI file, modify its timing, and determine which MIDI messages and events trigger changes in the Fusion parameter.

MIDI File

This browser control is used to specify the MIDI file that is used as the input for the modifier.

Time Scale

Time Scale is used to specify the relationship between time as the MIDI file defines it and time as Fusion defines it. A value of 1.0 plays the MIDI events at normal speed, 2.0 plays at double speed, and so on.

Time Offset

Time Offset adjusts the sync between the MIDI file's timing and Fusion's timing. If there is an unexpected delay, or if the MIDI file should start partway into or before some animation in Fusion, this control can be used to offset the MIDI data as required.

Result Offset, Result Scale

These sliders adjust the range of values produced by the modifier. By default, values between 0 and 1 (or -1 and 1 for PitchBend mode) are generated. This does not always suit the node/parameter, and scale can be used to make this range larger (such as * 0.0 - 2.0). Offset is used to provide some constant value as a base.

Result Curve

The Result Curve can also be used to adjust the output. However, this adjusts the curve of the result. By default, for any input MIDI data, the results fall linearly between 0.1 and 1.0 (for example, a velocity 127 note generates 1.0, whereas 63 generates approximately 0.5).

The Result Curve applies a gamma-like curve so that middle values can produce higher or lower results while still maintaining the full scale.

Mode

This menu provides Beat, Note, Control Change, Poly AfterTouch, Channel AfterTouch, or Pitch Bend, indicating from which MIDI event the values are being read. Beat mode is slightly different in that it produces regular pulses based on the tempo of the MIDI file (including any tempo maps).

The Beat mode does not use any specific messages; it bases its event timing on the tempo map contained in the MIDI file.

Combine Events

This menu selects what happens when multiple events occur at the same time. In Notes mode, this can happen easily. For other events, this can happen if Multiple Channels are selected.

Use this to take the result from the most recent event to occur, the oldest event still happening, the highest or lowest valued event, the average, sum, or the median of all events currently occurring.

Beat (Quarters) (Beat Mode Only)

This defines how often a beat occurs when in Beat mode. This is in quarter notes, so a value of 1.0 gives a beat every quarter.

Note Range (Note and Poly Aftertouch Modes Only)

This defines what range of notes causes a value to be generated. For example, use this to pick out the kick drum from a GM drum track by setting the note range between 35–36.

Pitch Scale (Note Mode Only)

Pitch Scale defines how much the result changes with pitch. A value of 1.0 causes the result to vary from 0.0 to 1.0 over the entire range.

Velocity Scale (Note Mode Only)

This defines how much the result changes with velocity. A value of 1.0 causes the result to vary from 0.0 to 1.0 over the entire range. This is added to the result from Pitch Scale for the final result.

Control Number (Control Change Mode Only)

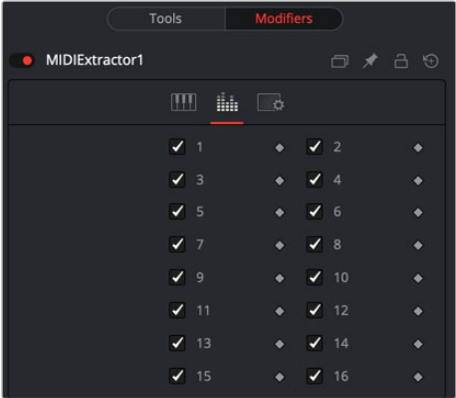
This specifies the MIDI controller number from which to extract events.

Envelope Controls (Note and Beat Modes Only)

These define an Envelope to follow for values before, during, and after the note or beat. Pre-Attack Time defines how long before the event it starts ramping up to the pre-attack level. Attack is the Time/Level to ramp to once the event has occurred, followed by the Decay ramp and Sustain, until the event stops. This stage is for Notes only. Beats have an instantaneous duration, so it goes straight to

Release. Release is the ramp-down time after the event finishes. When trying to do a Beat, set Release to some value, or there likely will not be much of a beat.

These values can be used to follow actual sounds in the MIDI sequence or just to create interesting effects. All time values used in the MIDI Extractor are in seconds.



The MIDI Extractor modifier Channels tab

Channels Tab

The Channels tab is used to select the Channels used in the modifier.

Channels

Channels checkboxes select which of the 16 channels in the MIDI file are actually considered for events. This is a good way to single out a specific instrument from an arrangement.

About MIDI

A single MIDI interface allows 16 channels. Typically, these are assigned to different instruments within a device or different devices. Usually, MIDI data is 7 bits, ranging from 0–127. In Fusion, this is represented as a value between 0–1 to be more consistent with how data is handled in Fusion.

There are numerous different MIDI messages and events, but the ones that are particularly useful with this modifier are detailed below.

MIDI Messages

Note On: This indicates that a note (on a specific channel) is being turned on, has a pitch (0–127, with middle C being 60) and a Velocity (0–127, representing how fast the key or strings or whatever was hit).

Note Off: This indicates that a note (on a specific channel) is being turned off, has a pitch (0–127, with middle C being 60) and a Velocity (0–127, representing how fast the key or strings or whatever was released).

Control Change: This message indicates that some controller has changed. There are 128 controllers (0–127), each of which has data from 0–127. Controllers are used to set parameters such as Volume, Pan, amount of Reverb or Chorus, and generic things like foot controllers or breath controllers.

MIDI Events

Channel Aftertouch: This event defines that pressure is being applied to the keys (or strings or whatever) during a note. This represents general, overall pressure for this channel, so it simply uses a pressure value (0–127).

Poly Aftertouch: This event defines that pressure is being applied to the keys (or strings or whatever) during a note. It is specific to each particular note and therefore contains a note number as well as a pressure value (0–127).

Pitch Bend

The Pitch Bend controller generally specifies the degree of pitch bending or variation applied to the note. Because pitch bend values are transmitted as a 14-bit values, this control has a range between -1 and 1 and a correspondingly finer degree of resolution.

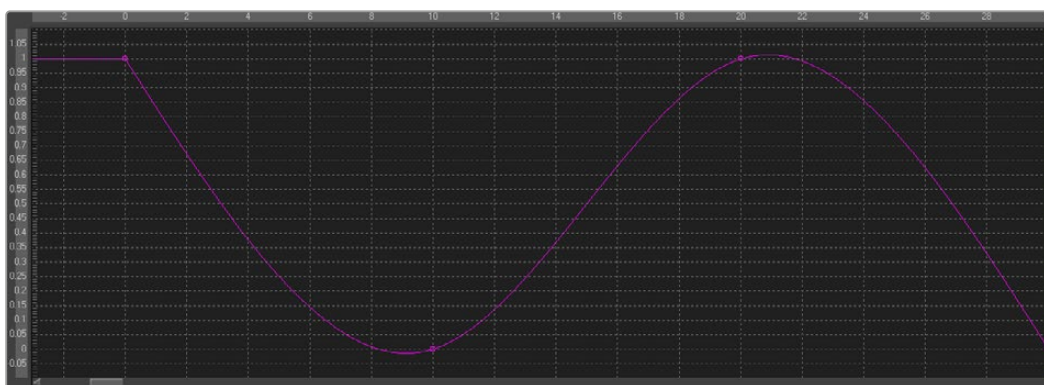
Natural Cubic Spline

The Natural Cubic Spline is one of the animation modifiers in Fusion and normally is applied to numerical values rather than point values. It can be applied by right-clicking a numerical control and selecting Modify With > Natural Cubic Spline.

NOTE: Unlike other spline types, Cubic splines have no control handles. They attempt to automatically provide a smooth curve through the control points.

Usage

Being an animation spline, this modifier has no actual Controls tab. However, its effect can be seen and influenced in the Spline Editor.



Natural Cubic Spline Editor

Offset (Angle, Distance, Position)

There are three Offset modifiers used to create variances between values. Depending on the modifier, these values relate to controls, paths, and points. The three types of Offset modifiers available in Fusion are:

- Offset Angle
- Offset Distance
- Offset Position

Offset Angle

The Offset Angle modifier outputs a value between 0 and 360 that is based on the angle between two positional controls. The Position and Offset parameters may be static, connected to other positional parameters, or connected to paths of their own. All offsets use the same set of controls, which behave differently depending on the offset type used. These controls are described below.

Offset Distance

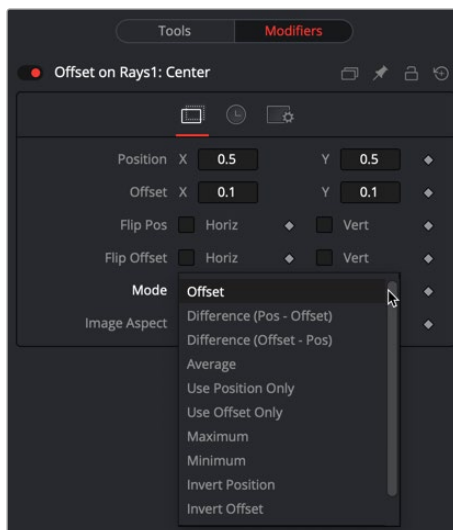
The Offset Distance modifier outputs a value that is based on the distance between two positional controls. This modifier is capable of outputting a value based on a mathematical expression applied to a position.

Offset Position

The Offset Position modifier generates a position (X and Y coordinates) that is based on the relationship between positional controls. This modifier is the equivalent of a calculation control except that it outputs X and Y coordinates instead of a value.

It can be applied by right-clicking a control and selecting Modify With > Offset.

Inspector



The Offset Position modifier controls

Offset Tab

The Inspector for all three Offset modifiers is identical. The Offset tab includes Position and Offset values as well as a Mode menu for selecting the mathematical operation performed by the offset control.

Position X and Y

The X and Y values are used by the Position to generate the calculation.

Offset X and Y

The X and Y values are used by the Offset to generate the calculation.

Flip Position Horizontal and Vertical

When these controls are selected, the Position is mirrored along the vertical or horizontal axis of the image.

Flip Offset Horizontal and Vertical

When these controls are selected, the Offset position is mirrored along the vertical or horizontal axis of the image.

Mode

The Mode menu includes mathematical operations performed by the Offset control. Available options include:

- Offset
- Difference (Position - Offset)
- Difference (Offset - Position)
- Average
- Use Position Only
- Use Offset Only
- Maximum
- Minimum
- Invert Position
- Invert Offset
- Invert Sugar
- Random Offset

Image Aspect

Adjust the modifier's output to compensate for the image aspect (not pixel aspect) of the project. A square image of 500 x 500 would use an Image Aspect value of 1, and a rectangular image of 500 x 1000 would use an Aspect Value of 2. The default value is always based on the current frame format selected in the preferences. To calculate image aspect, divide the width by the height. This control can also be used to create the illusion of aspect.



The Offset Time tab

Time Tab

Position Time Scale

This returns the value of the Position at the Time Scale specified (for example, 0.5 is the value at half the current frame time).

Position Time Offset

This returns the value of the Position at the Time Offset specified (for example, 10 is 10 frames back).

Offset Time Scale

This returns the value of the Offset at the Time Scale specified.

Offset Time Offset

This returns the value of the Offset at the Time Offset specified.

Example

This is a simple comp to illustrate one potential use of offsets.

- Create a new Comp 100 frames long.
- Create a node tree consisting of a black background and a Text node foreground connected to a Merge.
- In the Text Layout tab, use the Center X control to animate the text from the left side of the screen to the right.
- Move to frame 0.
- In the Text tab in the Inspector, right-click the Size control and select Modify With > Offset Distance from the contextual menu.
- This adds two onscreen controls: a crosshair for the position and an X control for the offset. These onscreen controls represent the Position and Offset controls displayed in the Modifiers tab.

The size of the text is now determined by the distance, or offset, between the two onscreen controls.

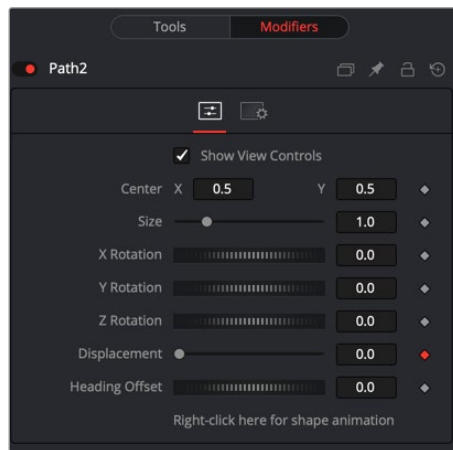
- Drag the X onscreen control in the viewer to see how the distance from the crosshair changes the size of the merge and by association the text.
- Both the crosshair and the X onscreen controls are animatable and can be connected to other controls.
- Position the X centered at the bottom of the viewer.
- In the Inspector, select the Modifiers tab.
- In the Offset on Text size section, right-click over Position and choose Connect To > PathConnect the position value of the Offset to the existing path by right-clicking the Position control and selecting Connect To > Path1 Position.
- Play the comp to view the animation.

Now, the text shrinks near the center of the path (when the distance between the offset and the path is at its minimum) and grows at its ends (where the distance between the offset and the path is at its maximum).

Path

The Path modifier uses two splines to control the animation of points: an onscreen motion path (spatial) and a Time spline visible in the Spline Editor (temporal). To animate an object's position control using a Path, right-click the Position control either in the Inspector or in the viewer and select Path from the contextual menu. This adds a keyframe at the current position. You can begin creating a path by moving the playhead and dragging the center position control in the viewer. The Spline Editor shows a displacement spline for editing the temporal value, or "acceleration," of the path.

Inspector



The Path modifier Controls tab

Controls Tab

The Controls tab for the path allows you to scale, reposition, and rotate the path. It also provides the Displacement parameter, allowing you to control the acceleration of an object attached to the path.

Center

The actual center of the path. This can be modified and animated as well to move the entire path around.

Size

The size of the path. Again, this allows for later modification of the animation.

X Y Z Rotation

The path can be rotated in all three dimensions to allow for sophisticated controls.

Displacement

Every motion path has an associated Displacement spline in the Spline Editor. The Displacement spline represents the position of the animated element along its path, represented as a value between 0.0 and 1.0. Displacement splines are used to control the speed or acceleration of an object's movement along the path.

To slow down, speed up, stop, or even reverse the motion of the control along the path, adjust the values of the points for the path's displacement in the Spline Editor or in the Inspector.

- A Displacement value of 0.0 in the Spline Editor indicates that the control is at the very beginning of a path.
- A value of 1.0 indicates that the control is positioned at the end of the path.
- Each locked point on the motion path in the viewer has an associated point on the Displacement spline.
- Unlocked points have a control point in the viewer but do not have a corresponding point on the Displacement spline.

Heading Offset

Connecting to the Heading adjusts the auto orientation of the object along the path. For instance, if a mask's angle is connected to the path's heading, the mask's angle will adjust to follow the angle of the path.

Right-Click Here for Shape Animation

It's possible to animate the shape of the path as well or to connect it to other path controls like Polyline Masks or Paint Strokes.

TIP: Switching Default Paths

You can change the default path type used when animating a position or center control to a path (if this is the preferred type of animation). Open the Preferences window and select the Global Settings. In the Default category, select the Point With menu and choose Path. The next time Animate is selected from a Position or Center control's contextual menu, a path is used.

Perturb

The Perturb modifier generates smoothly varying random values for a given parameter based on Perlin noise. It can be used to add jitter, shake, or wobble to any animatable parameter, even if the parameter is already animated. Its results are similar to those of the Shake modifier, although it uses a different set of controls that may be more intuitive. Unlike other random modifiers, you can apply the Perturb modifier to polylines, shapes, grid meshes, and even color gradients.

For example, to add camera shake to an existing path, right-click the crosshair and choose Insert > Perturb, and then adjust the Strength down to suit. Alternatively, right-clicking the path's "Right-click here for shape animation" label at the bottom of the Inspector lets you apply perturb to the path's polyline. This works best if the polyline has many points—for example, if it has been tracked or hand-drawn with the Draw Append pencil tool. A third usage option is to use the Insert contextual menu to insert the modifier onto the Displacement control. This causes the motion along the path to jitter back and forth without actually leaving the path.

NOTE: Perturb can only add jitter; it cannot smooth out existing animation curves.

Inspector



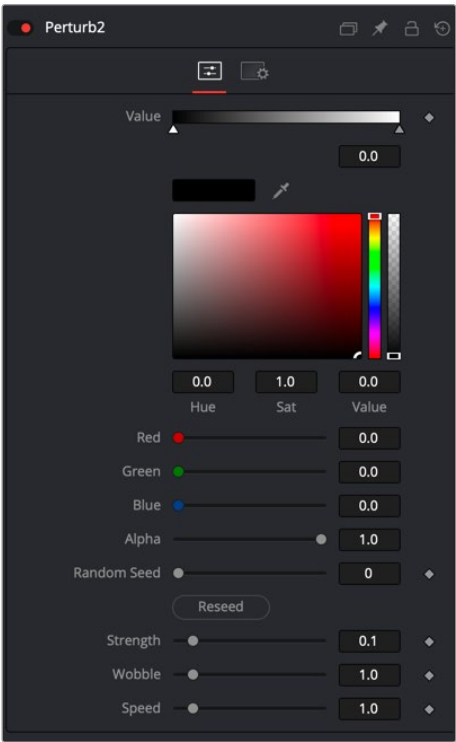
The Perturb modifier Controls tab

Controls Tab

The Controls tab for Perturb is mainly used for controlling the Strength, Wobble, and Speed parameters of the random jitter.

Value

The content of this control depends on what type of control the modifier was applied to. If the Perturb modifier was added to a basic Slider control, the Value is a slider. If it was added to a Gradient control, then a Gradient control is displayed here. Use the control to set the default, or center value, for the Perturb modifier to work on.



The Perturb modifier Gradient controls

Jaggedness

(Polylines and meshes only) This allows you to increase the amount of variation along the length of the polyline or mesh, rather than over time. Increasing Jaggedness gives a squigglier polyline or more tangled mesh, independent of its movement.

Phase

(Polylines and meshes only) Animating this can be used to move the ripple of a polyline or mesh along itself, from end to end. The effect can be most clearly seen when Speed is set to 0.0.

Random Seed Randomize

The Random Seed is used to “seed” the amount of jitter applied by the modifier. Two Perturb modifiers with identical settings, but different random seeds, produce two completely different results. Click the Randomize button to assign a random seed value.

Strength

Use this control to adjust the strength of the Perturb modifier’s output, or its maximum variation from the primary value specified above.

Wobble

Use the Wobble control to determine how smooth the resulting values are. Less wobble implies a smoother transition between values, while more wobble produces less predictable results.

Speed

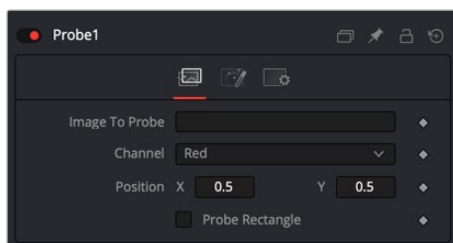
Increasing the Speed slider value speeds up the rate at which the value changes. This can increase the apparent wobbliness in a more predictable fashion than the Wobble control and make the jitter more frantic or languorous in nature.

Probe

The Probe modifier is one of the most versatile modifiers in Fusion. It allows you to control any numeric parameter by the color or luminosity of a specific pixel or rectangular region of an image. Think of driving the Brightness node by probing the pixel values of flickering lights in a shot, or measuring graded LUTs to compare values.

It can be applied by right-clicking a parameter and selecting Modify With > Probe.

Inspector



The Probe modifier Controls tab

Controls Tab

The Controls tab for the Probe modifier allows you to select the node to probe, define the channel used to drive the parameter, and control the size of the probed area.

Image to Probe

Drag a node from the Node Editor to populate this field and identify the image to probe.

Channel

Select the channel you want to probe. The usual options are:

- Red
- Green
- Blue
- Alpha

Luma

Once a Probe modifier is present somewhere in your comp, you can connect other node's values to its outputs as well. The Probe allows you to connect to its values individually:

- Result
- Red
- Green
- Blue
- Alpha

Position X Y

The position in the image from where the probe samples the values.

Probe Rectangle

By default, the Probe samples only the value of a single pixel at its position. By using the Probe Rectangle mode, you can sample from a larger area of pixels based on the Evaluation method.

Width Height Controls

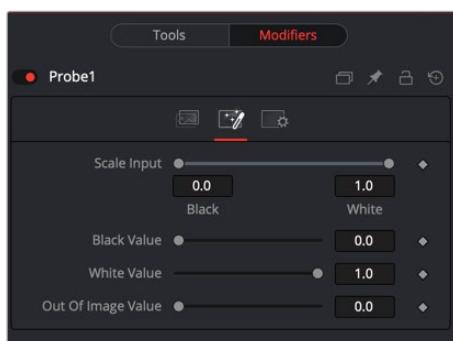
These determine the size of the area to be probed.

Evaluation

Sets how the pixels inside the rectangle are computed to generate the output value.

Options include:

- **Average:** All pixel values inside the rectangle are averaged.
- **Minimum:** The smallest value of all pixels inside the rectangle is used.
- **Maximum:** The highest value of all pixels inside the rectangle is used.



The Probe modifier Value tab

Value Tab

The Value tab controls the range or scale of the modifier adjustment, thereby adjusting the sensitivity of the Probe.

Scale Input

By default, the Probe generates the Black Value when the probed area results in a value of 0 (i.e., black), and it generates its White Value when the probed area results in a value of 1 (i.e., white). By using this range control, you can modify the sensitivity of the Probe.

Black Value

The value that is generated by the Probe if the probed area delivers the result set in Scale Input Black.

White Value

The value that is generated by the Probe if the probed area delivers the result set in Scale Input White.

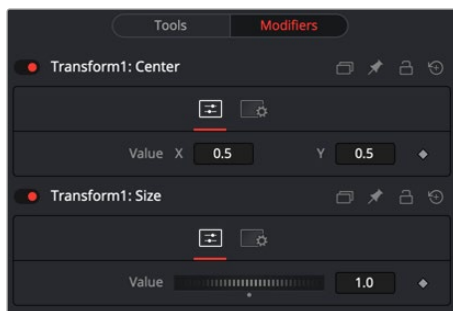
Out of Image Value

The value that is generated by the Probe if the probed area is outside the frame boundaries of the probed image. If probing a rectangle, this value does not generate before the entire rectangle is outside the frame boundaries of the image to be probed.

Publish

Only parameters that are animated will be available from the Connect To menu. To connect to non-animated parameters, you must Publish them first. Animated controls are automatically published, whereas static controls must be published manually.

To publish a static control, right-click the control and select Publish from the contextual menu.



The Publish modifier Controls tab

Controls Tab

The Controls tab shows the published control available for linking to other controls.

Published Value

The display of the published control is obviously dependent on which control is published from which node.

Resolve Parameter

The Resolve Parameter Modifier is used when creating a transition template in Fusion for use in DaVinci Resolve's Edit page or Cut page. When building a transition in Fusion, the Resolve Parameter modifier is added to any control you want to animate. The Resolve Parameter modifier automatically animates the parameter for the duration of the transition, allowing you to trim the transition in the Edit page or Cut page.

For example, to create a cross dissolve, do the following:

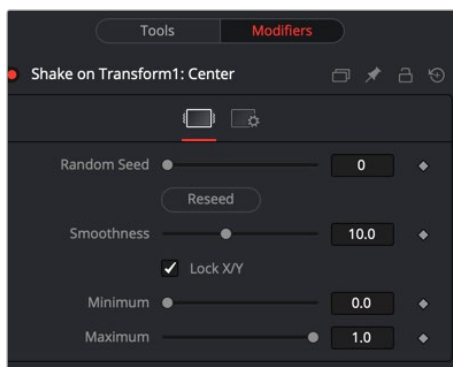
- 1 From the Effects Library, add a Fusion Composition to the Edit page Timeline.
- 2 In Fusion, add a Dissolve node to the Node Editor.
- 3 In the Inspector, right-click the Background/Foreground parameter, and then choose Resolve Parameter from the Modifier contextual menu. Adding the modifier to the Background/Foreground parameter automatically updates the slider if the transition is modified back on the Edit/Cut page.
- 4 In the Node Editor, right-click the Dissolve node and choose Macro > Create Macro.
- 5 When creating a macro that's to be used as a Fusion transition it's important that two inputs and one output are selected in the Macro Editor. In this example, under the Dissolve heading, enable the Output, Background and Foreground check boxes.
- 6 Give the transition a name, then save the macro from the top File menu. Save the macro in the following folder so it appears the Edit page or Cut page Effects Library:
For MacOS: \$TEMPLATE_MAC_OS_PATH/Transitions or \$TEMPLATE_MAC_USER_PATH/Transitions
For Windows: \$TEMPLATE_WIN_OS_PATH\Transitions or \$TEMPLATE_WIN_USER_PATH\Transitions
- 7 Quit and reopen DaVinci Resolve to update the list of transitions in the Effects Library.
- 8 On the Edit page, open the Effects Library. Navigate to Video Transitions Fusion Transitions, and the custom Fusion transition will be listed.

Shake

The Shake modifier is used to randomize a Position or Value control to create semi-random numeric inputs. The resulting shake can be entirely random. The motion can also be smoothed for a more gentle, organic feel.

To add the Shake modifier to a parameter, select Modify With > Shake from the parameter's contextual menu. The Shake modifier uses the following controls to achieve its effect. It can be applied by right-clicking a parameter and selecting Modify With > Shake.

Inspector



The Shake modifier Controls tab

Controls Tab

Random Seed

The Random Seed control contains the value used to seed the random number generator. Given the same seed, a random number generator always produces the same results. Change the seed if the results from the randomizer are not satisfying.

Smoothness

This control is used to smooth the overall randomness of the Shake. The higher the value, the smoother the motion appears. A value of zero generates completely random results, with no smoothing applied.

Lock X/Y

This checkbox is used to unlock the X- and Y-axis, revealing independent slider controls for each axis.

Minimum and Maximum

This control is used to determine the overall strength of the shake. The low values represent the lowest value that can be generated by the randomizer, and the high values represent the highest values. To create a Shake that moves a center crosshair anywhere within the image, set the Minimum to 0.0 and the Maximum to 1.0. To restrict the motion to a smaller shake in the bottom-right corner of the image, set the Minimum to 0.70 and the Maximum to 0.90.

Example

- Create a new comp, and then add and view a Text node.
- Type some text in the Text node.
- In the viewer, right-click over the Center control of the text and choose Modify With > Shake Position.
- In the Inspector, select the Modifiers tab and set the smoothing to 5.0.
- Set the Minimum to 0.1 and the Maximum to 0.9.

This adds some chaotic movement to the text. However, we can change this over Go to frame 0 and in the Inspector click the Keyframe button to the right of both the Minimum and the Maximum controls.

- Go to frame 0 and in the Inspector click the Keyframe button to the right of both the Minimum and the Maximum controls.
- Go to frame 90 and adjust the Minimum to 0.45 and the Maximum to 0.55.
- View the results.

Now, the text starts out by flying all over the screen and tightens in toward the center of the screen as the comp plays.

Track

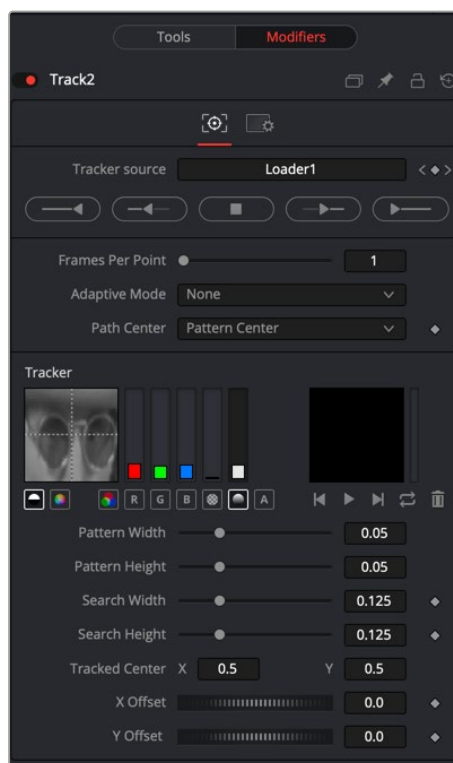
Although there is a standard Tracker node, you can also use a Tracker modifier to add a tracker directly to a parameter. To apply the Tracker modifier, in the viewer right-click the Center control of any transform, text, mask, or other positionable element. From the contextual menu, choose Object x Center > Modify With > Tracker.

Then choose one of three options:

- **Tracker Position:** Tracks a point from the source.
- **Steady Position:** Stabilizes based on a single point in the source.
- **Unsteady Position:** Adds original motion back after stabilizing.

This adds a modifier in the Inspector with a set of controls almost identical to those found in the Tracker node itself.

Inspector



The Tracker modifier

For an in-depth explanation of this node, see *Chapter 119, "Tracking Nodes,"* in the DaVinci Resolve Reference Manual, or Chapter 59 in the Fusion Reference Manual.

Tracker Modifier vs. Tracker Node

The differences between a Tracker modifier and a Tracker node are as follows:

- The Tracker modifier can only track a single pattern.
- The Tracker modifier can only output a single value and cannot be used for complex stabilization or match-moving procedures.
- The default source image for the modifier is the node immediately upstream of the node that contains the modifier (i.e., when adding a Tracker modifier to a Glow node with a Loader as its input, the Tracker Source input defaults to the output of the Loader). Set a different source image for the Tracker modifier by typing in the name of the node. Alternatively, drag the source node from the Node Editor into the Text Box control.

Example

Imagine that you need to track an actor's eyes so that an unearthly, alien glow can be applied to the eyes.

- Select the Loader node.
- Add a Glow node.
- Add an Ellipse mask to the Glow in the shape of one of the eyes.
- Right-click the center of that mask and select Modify With > Tracker > Position.

Since the track is on the mask, the tracker takes the glow as the image for tracking. This could cause problems since the eye might be very obscured by the glow. A cleaner source will be the Loader that feeds the glow.

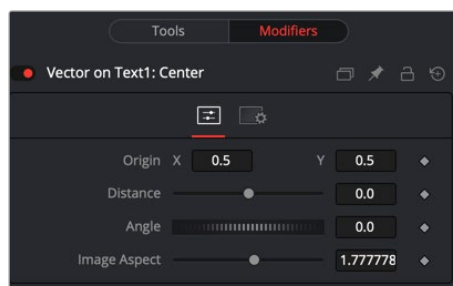
- Drag the Loader into the modifier Inspector's Track Source field.
- Track the actor's eye.
- Perform the same steps on the other eye.

Vector Result

The Vector Result modifier is used to offset positional controls, such as crosshairs, by distance and angle. These can be static or animated values.

It can be applied by right-clicking a control and selecting Modify With > Vector.

Inspector



The Vector Result modifier Controls tab

Controls Tab

Origin

This control is used to represent the position from which the vector's distance and angle values originate.

Distance

This slider control is used to determine the distance of the vector from the origin.

Angle

This thumbwheel control is used to determine the angle of the vector relative to the origin.

Image Aspect

This slider control is used primarily to compensate for image aspect differences. A square image of 500 x 500 would use an Image Aspect value of 1, while a rectangular image of 500 x 1000 would use an Image Aspect value of 2. The default for this value is taken from the current Frame Format preferences using width/height. It may be necessary to modify this control to match the current image.

Example

- Create a 100-frame comp.
- Create a simple node tree consisting of a black background and a Text node foreground connected to a Merge.
- Enter some text in the Text node.
- Select the Merge node.
- In the viewer, right-click the Center control of the Merge and choose Modify With > Vector Result.

This adds a crosshair onscreen control for the Vector distance and angle. The onscreen control represents the Distance and Angle controls displayed in the Modifiers tab.

- In the Modifiers tab of the Inspector, drag the Distance control to distance the text from the Vector origin.
- Drag the Angle thumbwheel to rotate the text around the Vector origin.
- This is different from changing a pivot point, since the text itself is not rotating.
- These points are animatable and can be connected to other controls.
- In the Inspector, right-click the Origin control and choose a path to add a motion path modifier to the Origin control.
- Verify that the current frame is set to frame 0 (zero) and use the Origin controls in the Inspector or drag the Vector Origin crosshair to the bottom-left corner of the screen.
- On the Vector Angle thumbwheel, click the Keyframe button to animate this control.
- Set the Angle thumbwheel to a value of 10.
- Go to frame 100 and click at the top-left corner of the screen to move the Vector Origin crosshair.
- Set the Vector Angle thumbwheel to a value of 1000.
- Play the comp to see the results.

This causes the text to orbit around the path just created.

XY Path

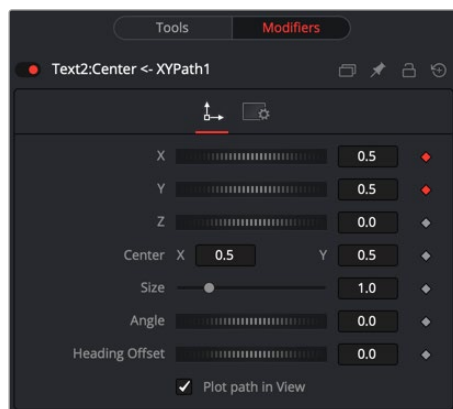
The XY Path type uses two separate splines for the position along the X-axis and for the position along the Y-axis.

To animate a coordinate control using an XY path, right-click the control and select **Modify With > XY Path** from the contextual menu.

At first glance, XY paths work like Displacement paths. To describe the path, change frames and position the control where it should be on that frame, and then change frames again and move the control to its new position. Fusion automatically interpolates between the points. The difference is that no keyframes are created on the onscreen path.

Look in the Spline Editor to find the X and Y channel splines. Changes to the control's values are keyframed on these splines. The advantage to the XY path is that it becomes very easy to work with motion along an individual axis.

Inspector



XY Path modifier controls

X Y Z Values

These reflect the position of the animated control using X, Y, and Z values.

Center

The actual center of the path. This can be modified and animated as well to move the entire path around.

Size

The size of the path. Again, this allows for later modification of the animation.

Angle

The angle of the path. Again, this allows for later modification of the animation.

Heading Offset

If another control (for example, a mask's Angle) is connected to the path's heading, this control allows for adding or subtracting from the calculated angle.

Plot Path in View

Toggles whether or not the actual path is displayed in the views.

Switching Default Paths

You can change the Default Path type to XY path (if this is the preferred type of animation). Open the Default category in the Global Preferences and locate the Point With drop-down menu. Change this from the current value to XY Path. The next time Animate is selected from a Coordinate control's contextual menu, an XY path is used instead of a Displacement path.



Color

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